Earth and Planetary Science Letters, 103 (1991) 365–378
Elsevier Science Publishers B.V., Amsterdam

[XLeP]

Paleomagnetism in the Tajikistan: continental shortening of European margin in the Pamirs during Indian Eurasian collision

Jean-Pierre Pozzi and Hugues Feinberg

École Normale Supérieure, Département de Géologie. 24, rue Lhomond 75231 Paris Cedex 05, France

Received July 26, 1990; revised version accepted November 13, 1990

ABSTRACT

A stable and well defined magnetization with $D = 350°$, $I = 55.5°$ and $\alpha_{95} = 4°$, carried by hematite, is isolated in the red beds of the Tajik basin. The fold test is positive (99% probability) and this magnetization predates the Late Eocene–Early Oligocene folding phase in the Tajik basin. The coordinates of the VGP are: 82°N, 323°E, dp = 4, dm = 6 with a paleolatitude of Dushanbe (38°38N, 68°51E) of 35.6 ± 5°.

A high blocking temperature component with weak intensity and reverse polarity has been demonstrated in the site of Ragoon, but it has not been isolated. This component predates the Long Cretaceous Normal Period and is compatible with the older stratigraphic age attributed to the formation (Valanginian); these arguments indicate that this component is likely of primary origin. The other component, of higher intensity and clearly isolated in all the sites, is a pre-folding remagnetization. The process of remagnetization is attributed to the tectonic events, such as fluids migration and slight heating during a few million years, which took place just before the first Late Eocene–Early Oligocene phase of folding

An anticlockwise rotation of 26° encountered by the region of Dushanbe is in good agreement with the indentation model of collision leading to anticlockwise rotation West of the Indian wedge and clockwise rotation to the east. A comparison of the paleolatitude of Dushanbe, computed from the direction of magnetization, and the paleolatitude expected from the apparent polar wander path for Eurasia, gives an estimate of the post-collision continental shortening north of the Tajik basin. This shortening did not exceed a few hundred kilometers and was probably between 100 and 300 km. An estimate of the roles of the different classical mechanisms of crustal shortening is discussed and a paleogeographic reconstruction giving the location of the southern margin of Eurasia at 45 Ma is proposed. The paleolatitude deduced from the prefolding remagnetization of the Red Beds of the Tajik basin is then compared with the results obtained by Soviet workers in Tertiary sediments of the same area, revealing strong contradictions in interpretation. This contradiction emphasizes the problem of the rigidity of European plate.

1. Introduction

The Indian–Eurasian collision has been intensively investigated during the last decade. Paleomagnetism has contributed to the reconstruction of the timing and of the geometry of the collision through rotational movements and paleolatitudes determinations. In the central and eastern part of the Himalayan belt, the southern margin of Asia has been sampled for paleomagnetism during a three year French–Chinese program of cooperation (1980–1983). The precollision paleolatitude of the Cretaceous sediments of the Lhasa microcontinent, southern Tibet, (North of the Tsang Bo suture zone), has been found to be between 20°N [1,2] and 14°N [3]. Sampling in the Tsang Po suture zone and just south of the suture, near Dingri, (southern Tibet), on the northern edge of the Indian subcontinent, has permitted a separate estimate of the convergence in the suture zone and in the intracontinental thrusts [4,5].

In the Western part of the Himalayan belt, the Hindu Kush–Pamir–Karakorum syntaxial bend has been shown by paleomagnetism to be a secondary tectonic feature according to the indentation model of collision [6,7]. However, few paleomagnetic studies have been devoted to a precise determination of the precollision paleolatitude of the southern margin of Eurasia. Results from Cretaceous sediments have been reported by Burtman and Gurariy [8] and by Bazhenov and Burtman [9], but these results were not taken into account in the reconstructions of Klootwijk [10] and Besse [11]. The aim of this paleomagnetic

0012-821X/91/$03.50 © 1991 – Elsevier Science Publishers B.V.

study of Cretaceous sediments of the Tajik basin is to determine the precollision paleolatitude of the southern margin of Eurasia and to deduce the continental shortening which resulted from the indentation of Asia by the Indian subcontinent.

2. Geological setting

Near the western end of the Himalayan belt, large-scale Cenozoic tectonics are satisfactorily explained by a rigid wedge penetrating into a plastic medium [7]. The wedge corresponds to the northwestern part of India, south of the Chaman Fault to the west and south of the Karakorum Fault to the east. This hypothesis is supported by paleomagnetic results [10], which indicate large rotations of opposite dextral and sinistral senses in the eastern and western limbs of the syntaxis.

The present geometry of the Tajik basin, which pinches out to the west between the Pamir and the Tien Shan (Fig. 1), is the result of the shortening of the strongly folded Mesozoic sediments of the Peter the First and Alai ranges. Key features of recent tectonics in the Tajik basin include distributed folds with a westward vergence and a general arc shape of the folding from NE–SW in the region of sampling (near Dushanbe, Fig. 1), to a NW–SE direction south of the Afghan border. The Tajik depression is filled with up to 4 km of platform sediments. The Mesozoic and Cenozoic sediments show locally intense folding and penetrative deformation [12]. The Lower Cretaceous red beds constitute a succession of thick formations with various tectonic dips and seem very suitable for a paleomagnetic study. A recent paleomagnetic study in the same area [13], has shown that the magnetization of the Lower Cretaceous red beds is a pre-folding remagnetization. The proposed palinspastic reconstruction of the South Tien Shan and Tajik depression was based only on rotation angles of the different regions with a position of the Tajik basin far to the north.

3. Sampling and measurements

The Tajik depression was sampled for the study of the geomagnetic anomalies related to earthquakes. Carefully oriented blocks of both sediments and volcanics (andesites), where taken. Cretaceous red beds were sampled at four sites,

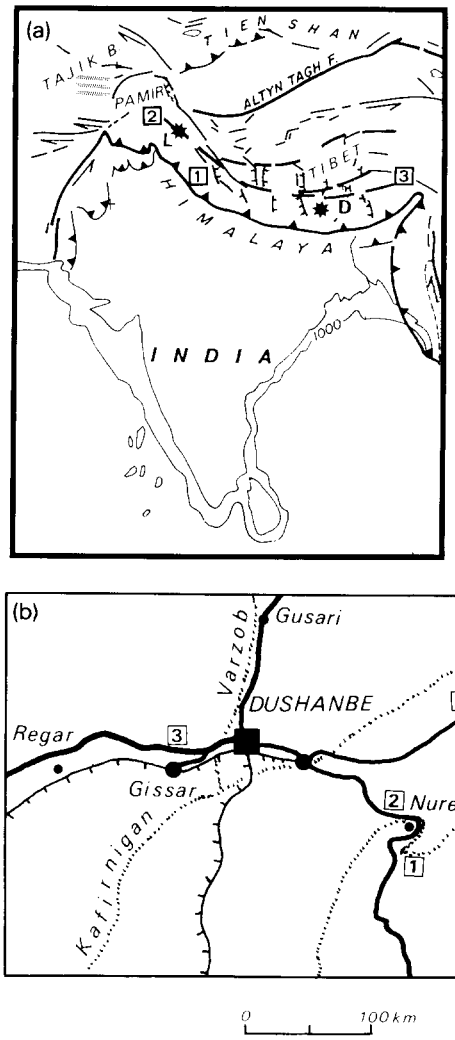


Fig. 1. (a) Structural sketch map: *1*, Himalayan Frontal Thrust; *2*, Indus suture; *3*, Zang Po suture; stars with *L* and *D* correspond respectively to the Ladakh and to Dingri, the studied zone is shaded. (b) Location of the sites of sampling: *1*, Nurek Dam; *2*, Nurek Pass; *3*, Gissar; *4*, Ragoon.

(Fig. 1): (1) in fresh layers excavated for the Nurek Dam—site Nurek Dam (specimens 11-01 to 11-16); (2) in outcrops of the Nurek road—site Nurek Pass (specimens 12-01 to 12-09); (3) in the Gissar ranges—site Gissar (specimens 14-01 to 14-12.); and (4) near the dam of Ragoon—site Ragoon. (specimens 13-01 to 13-16.) The sampled formations are dated from Valanginian to Aptian (14).

The susceptibility and the intensity of natural remanent magnetization have been used to compute the seismomagnetic effects in the region [15].

Recently, the samples of red sandstones were demagnetized by step heating to determine the characteristic directions of magnetization. Measurements of the direction and of the intensity of magnetization were made using a cryogenic magnetometer.

The magnetic minerals were investigated in three specimens from each site, using: (a) acquisition curves of Isothermal Remanent Magnetizations (IRM) in induction up to 2.1 T (IRM_{max}); (b) continuous thermal demagnetization of IRM_{max} using a fully automatic magnetometer designed in the laboratory; (c) evolution of the magnetic susceptibility during continuous heating; and (d) changes in magnetic viscosity during continuous heating.

4. Results

4.1. Magnetic minerals

One example of an IRM acquisition curve for each site is shown in Figs. 2a, 3a, 4a and 5a. The curves are characteristic of a high-coercivity mineral. The coercivity of the remanence ranges from 0.3 to 0.5 T. The continuous thermal demagnetization curves of IRM_{max} are shown in Figs. 2b, 3b, 4b and 5b, for one specimen from each site. They show various blocking temperatures and Curie temperature of hematite. The susceptibility and the viscosity measured during continuous heating do not reflect mineralogical changes. In conclusion, hematite is likely to be the sole carrier of the remanent magnetization.

4.2. Characteristic directions of remanent magnetization

The direction and intensity of the natural remanent magnetization have been followed during step-heating demagnetization. The results are reported site by site.

Site Nurek Dam. The mean intensity of the remanent magnetization is 7×10^{-2} A/m. Examples of Zijderveld vector demagnetization diagram are shown in Fig. 6. The characteristic directions of

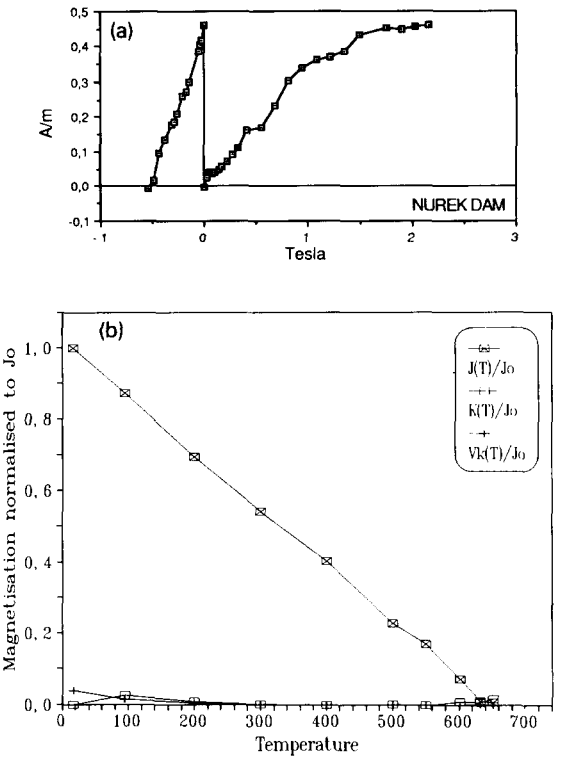


Fig. 2. Site Nurek Dam. (a) IRM acquisition curve and coercivity of the remanence. (b) High-temperature continuous measurements (relative value). $J(T)$ = continuous thermal demagnetization of IRM_{max}; $K(T)$ = initial susceptibility; and $Vk(T)$ = magnetic viscosity of induced magnetization defined as the magnetization acquired from 30 s to 7 min after the application of a field of 0.3 mT.

magnetization have been computed using a modified Kirchvink program [16]; they are as follows:

Before tectonic correction

N	lat.	long.	D	I
16	38.24	69.15	346.6	26.0

After tectonic correction *

D	I	α_{95}	k
324.2	55.9	4.8	53

* strike 106, dip 38.

N is the number of specimens; lat. and long. are the geographic coordinates of the sampling site; D, I, α_{95} and k define the characteristic direction of the remanence.

Site Nurek Pass. This site was sampled 30 km from Nurek Dam on the right hand side of the

road. Samples from the two sites do not differ in lithology. The mean intensity of natural remanent magnetization is 5×10^{-2} A/m.

Examples of vector diagram demagnetization are given in Fig. 7. A characteristic component of magnetization is isolated in the following direction:

Before tectonic correction.

N	lat.	long.	*D*	*I*
9	38.26	68.95	172.9	36

After tectonic correction *

D	*I*	α_{95}	*k*
355.5	54	12.4	14

* strike 84, dip 88.

Site Gissar. The mean intensity of NRM is 8×10^{-2} A/m. Examples of demagnetization vector diagram are shown in Fig. 8. Only one component

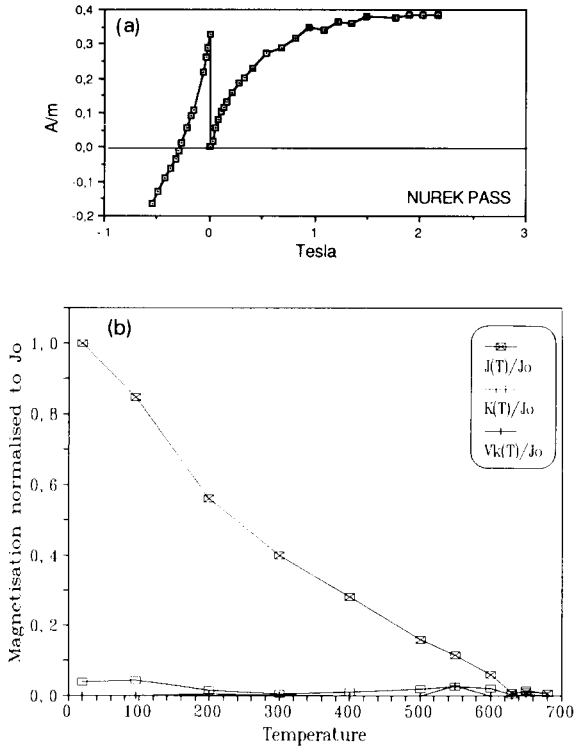


Fig. 3. Site Nurek Pass, same convention as Fig. 2.

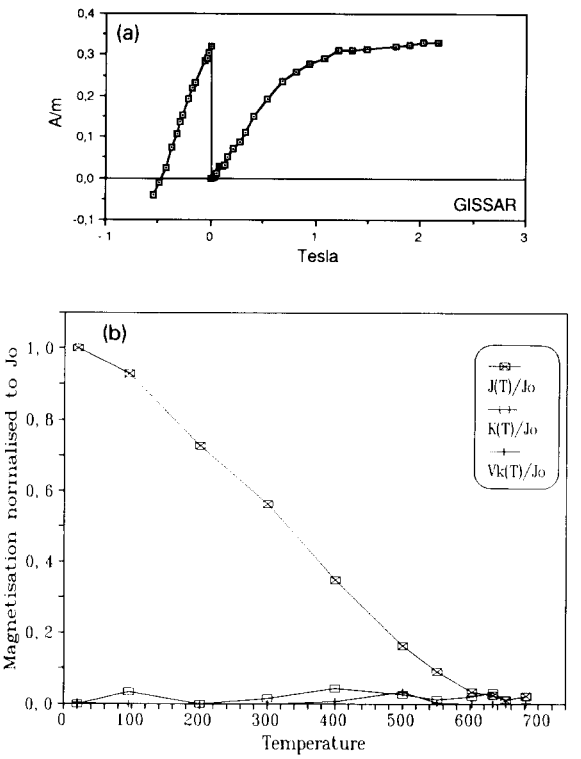


Fig. 4. Site Gissar, same convention as Fig. 2.

is revealed. The characteristic direction of magnetization is:

Before tectonic correction

N	lat.	long.	*D*	*I*
12	38.30	68.36	43.6	52.4

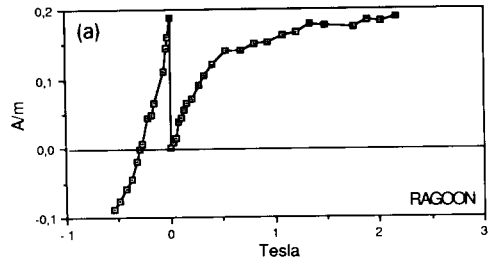
After tectonic correction *

D	*I*	α_{95}	*k*
6.3	43.7	5.2	61

* strike 219, dip 33.

Site Ragoon. Specimens from Ragoon have a mean NRM intensity of 3×10^{-2} A/m, lower than the NRM measured in the other sites. Moreover, the behaviour of the specimens during thermal demagnetization is characteristic and very interesting. Examples of Zijderveld diagrams are shown in Fig. 9. A component of magnetization of weak intensity can be seen after demagnetization at high temperature.

On stereographic plots the demagnetization paths of all the specimens form circles of remagnetization which evidence an evolution which begins after 550°C and continues toward a stable



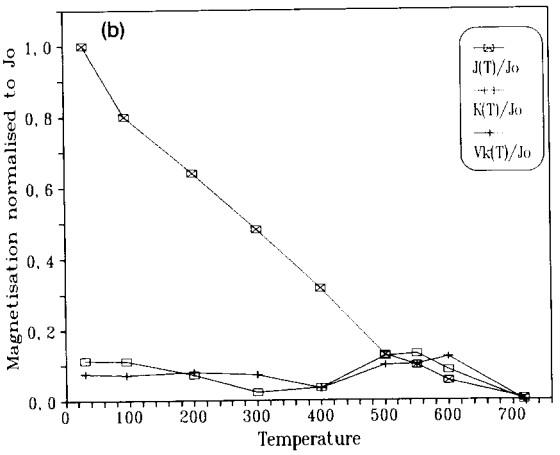


Fig. 5. Site Ragoon, same convention as Fig. 2.

end point which was not reached. The corresponding magnetization (component A) is of low intensity and of reverse polarity. Four examples of remagnetization circles are represented in Fig. 10. The intersections of the 16 circles corresponding to all the specimens, are not clearly defined. The direction of the component B, of blocking temperatures up to 550°C, is as follows:

Before tectonic correction

N	lat.	long.	D	I
16	38.50	69.00	336.4	0.2

After tectonic correction *

D	I	α_{95}	k
359.1	60.4	3.7	89

* strike 47, dip 67.

4.3. *Result of the fold test*

The directions of magnetization for the four sites are represented in Fig. 11 before and after tectonic correction. The directions of magnetization computed from all the samples are as follows:

Before tectonic corrections

N	D	I	α_{95}	k_1
53	351.9	38.3	11.5	3

After tectonic corrections

N	D	I	α_{95}	k_2
53	350.5	55.4	4.0	23

The precision parameter k increases from 3 to 23 (for $N = 53$ entries), which indicates that the results pass the fold test at a significant level with better than 99% confidence [17]. Consequently, the magnetization of the Red Formation from the Tajik basin was acquired before the Late Eocene–Early Oligocene (about 38 Ma) folding.

The results are summarized in Table 1.

5. Discussion

The prefolding components of magnetization, i.e. the characteristic remanence of Nurek Dam,

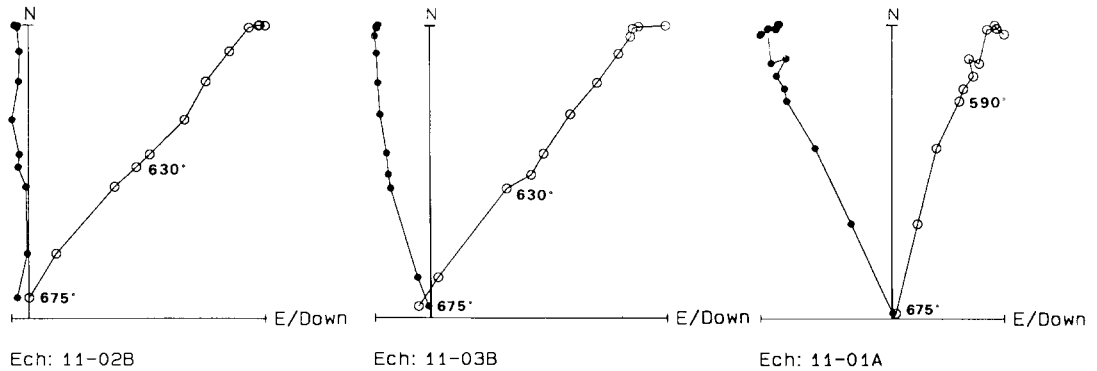


Fig. 6. Site Nurek Dam. Examples of vector demagnetization diagram during thermal treatment. Full (open) circles correspond to projections onto the horizontal (vertical) plane.

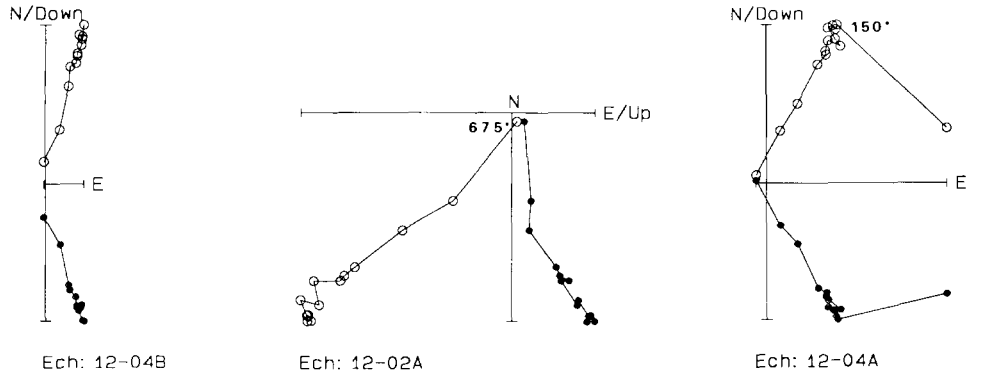


Fig. 7. Site Nurek Pass, same convention as Fig. 6.

Nurek Pass and Gissar sites, and the component B of Ragoon were acquired during the time between the sedimentation of the sandstones and the first phases of folding in the Tajik Basin dated Upper Eocene–Early Oligocene.

Our interpretation is that the component A, of reverse polarity, evidenced at the Ragoon site is a primary component acquired during the time interval between the older stratigraphic age of the sediments (Valanginian, 130 Ma), and the end of

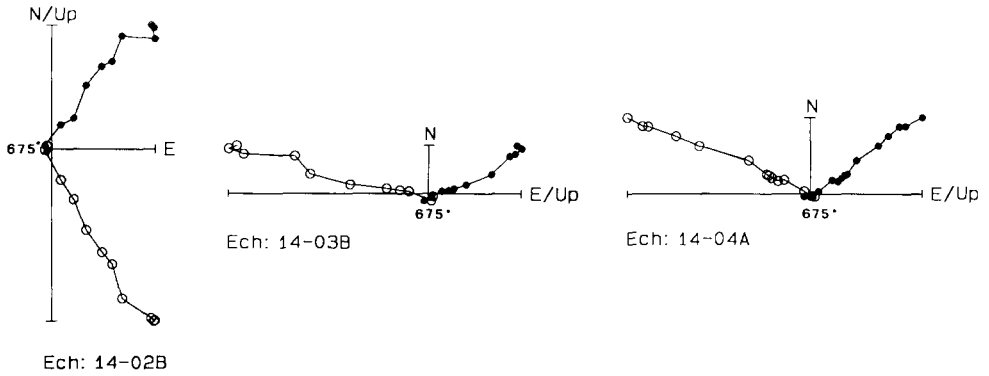


Fig. 8. Site Gissar, same convention as Fig. 6.

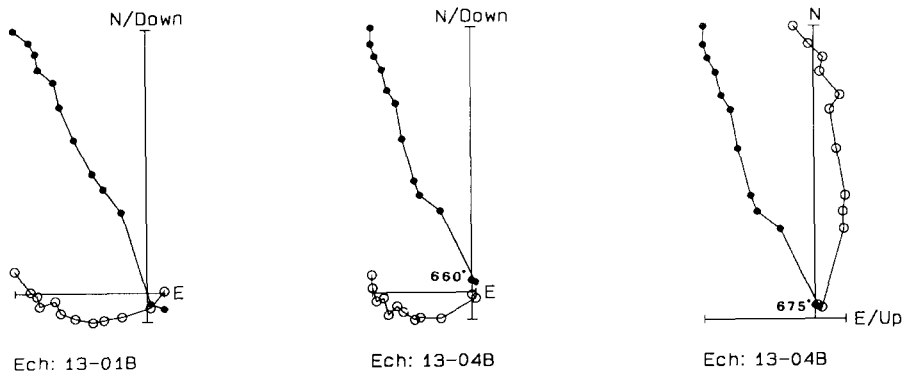


Fig. 9. Site Ragoon, same convention as Fig. 6.

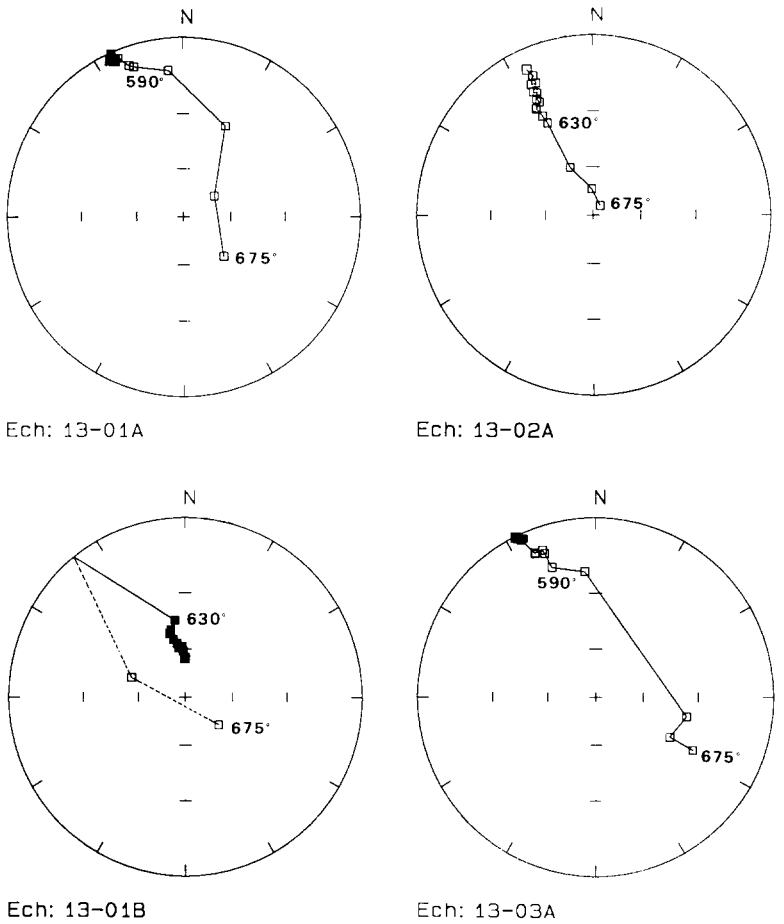


Fig. 10. Examples of remagnetization circles for the site of Ragoon. Stereographic projection in geographic coordinates except for the sample 13-01B which is in stratigraphic coordinates.

the *Chron Mo*, before the Long Cretaceous Normal Superchron.

The estimation of the time elapsed between the acquisition of the components A and B is the key problem of the discussion.

Component B of Ragoon probably represents the characteristic direction of magnetization found at the Nurek Dam, Nurek Pass and Gissar sites where it has completely overprinted the original component A; it is likely that a significant tectonic

Table 1

Results in the area of Dushanbe

Site	N	Lat.	Long.	Before unfolding				After unfolding			
				D	I	k	α_{95}	D	I	k	α_{95}
Nurek Dam	16	38.2	69.2	346.6	26	53	4.8	324.2	55.9	53	4.8
Nurek Pass	9	38.3	69	172.9	36	14	12.4	355.5	54	14	12.4
Gissar	12	38.3	68.4	43.6	52.4	61	5.2	6.3	43.7	61	5.2
Ragoon	16	38.5	69	336.4	0.2	89	3.7	359.1	60.4	89	3.7
Mean	53			351.9	38.3	3	11.5	350.5	55.4	23	4

Paleolatitude computed in Dushanbe (38°38N 61°58E) = 36° N.

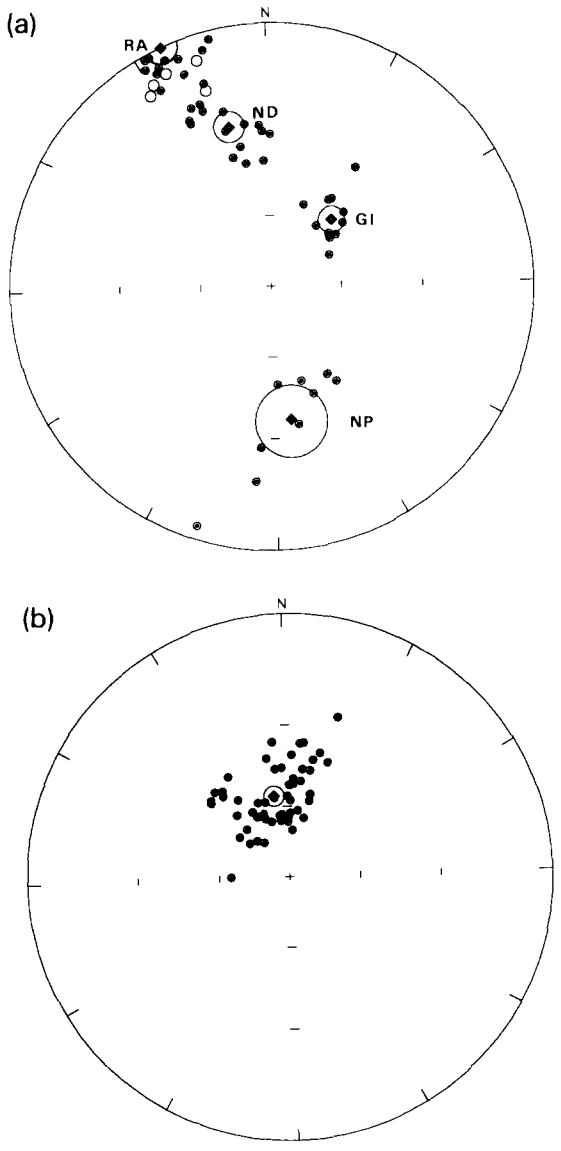


Fig. 11. Fold test: directions of magnetization before (a) and after (b) tectonic correction in stereographic projection.

event was necessary to create the conditions for fluids circulation necessary for widespread remagnetization. It is tempting to correlate this event with the collision of India and the southern Margin of Eurasia which started between M 22 and M 23 (50 Ma) [11,18,19,20], but only became effective in the Tajik basin at about 40 Ma after the contact of the continental margins [19,21]. This is the only major tectonic event in the Tajik basin since Valanginian time. Consequently we must assume that the age of the component B of remagnetization is from 45 Ma to 38 Ma, taking into account that the subsurface flow phenomena, which is likely responsible for the remagnetization, must have been active before the collision and folding.

In this case of a close relationship between a major tectonic event and a remagnetization we have to ensure that the magnetization has not been acquired during folding. In fact, the declination of the Nurek Dam site (324°) deviates from the mean declination (350°) after tectonic correction. The directions do not overlap during unfolding and the correlation of declination and structural trends (strikes) is negative following Perroud [22]. There is no reason to suspect that the magnetization was acquired during the folding. After structural correction, the Nurek Dam site shows the same inclination as the other sites suggesting the effect of plunging folds which is difficult to take into account in this complex structural setting.

The paleomagnetic pole computed in Dushanbe (38°38N, 68°51E) is:

	lat.	long.	dp	dm
VGP	82	323	4	6

The paleolatitude of Dushanbe is 36° ± 5°.

5.1. Displacements in latitude

The displacement of the Dushanbe area with respect to Eurasia is found by comparing the paleolatitude of 36°N at 45 to 38 Ma, with the paleolatitude deduced from apparent polar wander path of cratonic Eurasia. For that comparison two sets of synthetic apparent polar wander paths were used, one by Besse [11], the other one by Westphal et al. [23], though other reliable curves are available [24]. For the period we are concerned with, the poles proposed by Besse [11] and Westphal et al. [23] for Europe are presented in Table 2.

The paleolatitude versus time diagram, computed from APWP of Europe [11] is shown in Fig. 12, and is used for the determination of continental shortening. For a magnetization acquired after 45 Ma and before 38 Ma, an estimate of the north–south component of the continental displacement is 100 km (Fig. 12). When error bounds are included for the paleolatitudes of both magne-

tization and the kinematic history of Europe this shortening is found to range between 0 and 600 km (which implies an unlikely summation of error bars). Using the apparent polar wander path proposed in [23], estimates of the continental shortening range between 350 and 400 km respectively for a time of acquisition of magnetization of 45 and 38 Ma.

These two determinations are coherent and lead to a probable amplitude of continental shortening of a few hundred kilometers.

5.2. Relative rotation

The expected declination, computed from the poles [11], is as follows:

Age	Pole lat.	Pole long.	Expected decl.	d*D*	relative rotation
45	77	162	16.2	4.8	26 ± 9

The rotation of the sampled area of the Tajik basin is $26° \pm 9°$ anticlockwise. This result is in good agreement with the hypothesis of an indentation of Eurasia by Indian subcontinent. Paleomagnetic results obtained from Indo-Pakistan and reviewed in [10], show a compatible pattern of rotations, anticlockwise west of the indenter and of opposite sense on the other limb of the syntaxis.

5.3. Paleogeographic reconstruction (Fig. 13)

The paleolatitude of the Tajik basin along with the interpretation of a late pre-folding remagnetization of the red beds constrains the position of the Western part of the asiatic margin and allows us to propose a paleogeographic reconstruction of the margin at 45 Ma after the initial collision, but before the contact of the margins in the western part of the Himalayas.

The other data taken into account to constrain the paleogeographic reconstruction are the Ladakh Intrusives, now situated just north of the Indus suture zone. One of the components of magnetization isolated in the formation gives a paleolatitude of 10° N [19]. This component is related to the initial collision of Indian continent with the mass of the Ladakh island arc at pre-middle Eocene times (50 Ma). Between this initial collision and the contact of Indian and Eurasian continental margins, India (with the Ladakh island arc stet to it) moved northward. The consumption of the former oceanic crust took place along one or more subduction zone [25,26].

Farther east, in South Tibet, the Lingzizong volcanics yielded a paleolatitude of about 12° N, deduced from a magnetization acquired between 60 and 45 Ma [2,3]. At about the same longitude, on the northern margin of the Indian continent, the magnetization of the Dingri limestones, now immediately south of the suture, constrains the paleolatitude of the margin to 6° N at 50 Ma [4]. This margin has subsequently suffered deformations related to the different phases of the collision which have occurred between 50 Ma and 35 Ma [4]. The relative position and the paleolatitude

Table 2

Expected paleolatitudes deduced from two sets of synthetic polar wanderpaths

Age		Lat.	Long.	A_{95}	Expected paleolat. at Dushanbe	dp
Besse [11]						
Ku	80	75	195	3.8	28.7	3.2
Ku	70	75	185	5.0	30.6	4.4
Tpa	60	78	167	3.5	35.6	3.5
Te	50	77	162	2.9	36.4	4.0
To	40	77	162	3.9	36.4	4.0
To	30	78	130	3.9	43.2	4.7
Westphal et al. [23]						
Ku	80	78	188	12.0	31.8	11
Ku–Tpa	65	78	163	7.0	36.4	7.2
Tpa–Te	60	78	156	7.0	37.9	7.5
To	35	83	136	9.0	40.8	10.3

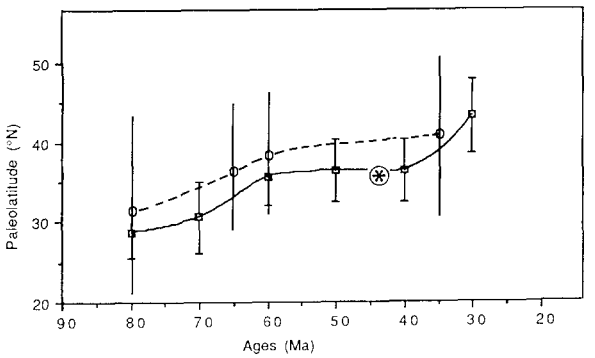


Fig. 12. Paleolatitude versus time diagram; open square after [11] and open circle after (23). Asterisk = paleolatitude computed from the pole of Dushanbe (this paper).

of the Lingzizong formation and of the Dingri limestones at 45 Ma are dependent on the timing of these deformations. In the hypothetical case of a rigid collision, the two formations of Lingzizong and Dingri should have moved northward, together with India, at the same velocity. Between

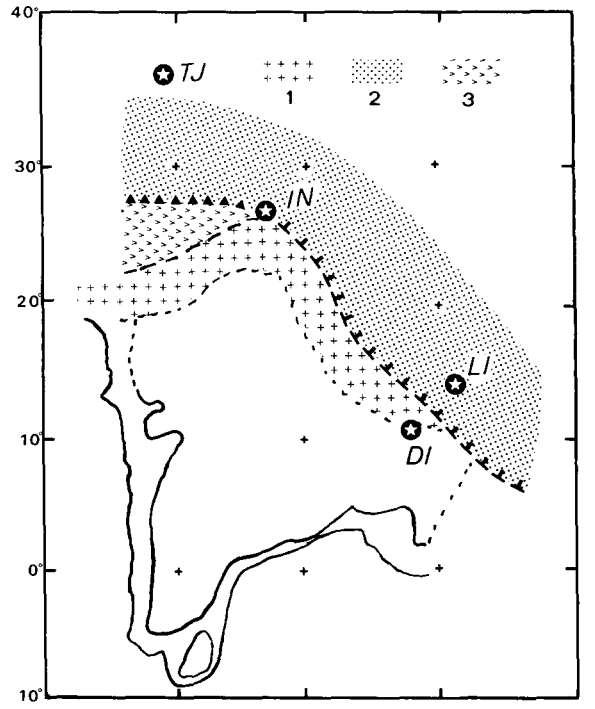


Fig. 13. Paleogeographic reconstruction at 45 Ma. Circled stars correspond to paleolatitudinal constraints: *IN* = computed from APWP of India; *TJ* = from red beds of the Tajik basin; *LZ* = from the Linzizong formation; *DI* = from Paleocene limestones of Dingri. *1*, Indian margin; *2*, Eurasian margin; *3*, Marginal basin.

50 Ma, the age of the collision in South Tibet (supposed to be the age of the magnetizations), and 45 Ma, they would have moved about 7° in latitude. Before the suture became blocked (45 Ma) at the longitude of Dingri [4,27], about 500 km (±500 km) of continental crust was subducted in the suture [4]. Consequently 14° N is a reasonable approximation for the paleolatitude of South Tibet at 45 Ma which accounts for part of this shortening. When the suture became blocked, the motion was transferred to the MCT and much of the 400 km (±400 km) of the convergence occurred by thrusting. A paleolatitude of 11° N is a reasonable estimate for the paleoposition of Dingri at 45 Ma. The position of India at 45 Ma has been deduced from the plate reconstructions of Patriat et al. [21].

The APWP of India has been improved recently [11], and one may obtain an estimate of the paleoposition of the southern continental margin of Asia at the longitude of the Ladakh by the paleoposition of the Northern continental margin of India (Ladakh Intrusives) at the time of the intimate collision, 38 Ma. This estimate leads to a paleolatitude of 27° N. (the coordinates of the pole used in this determination is: long. = 309, lat. = 82, $A_{95} = 3.9$ [11]).

These constraints lead to the paleogeographic reconstruction of Fig. 13. It appears that the paleolatitude of the Ladakh Intrusives is not reconcilable with the paleolatitude of the Tajik basin and that a wide oceanic marginal basin has been absorbed North of the suture in the western part since the initial collision.

5.4. Shortening of Eurasian margin

The convergence of the margins of India and Eurasia can be accounted for by four classical mechanisms: (1) closure of marginal basins; (2) crustal thickening through folding and large scale thrusting; (3) underthrusting of continental crust; and (4) lateral extrusion of lithospheric blocks out of the way of the course of India. The role of each process can be evaluated in the western part of the collision and compared with the eastern part in South Tibet.

The comparison of the paleolatitude found for the Tajik basin and the paleoposition of the northern margin of India (or southern margin of Europe), at the time of the intimate collision gives

an estimate of the continental shortening of the margin south of the Tajik basin of about 1000 km by folding, thrusting and lateral expulsion of lithospheric blocks.

The paleolatitude of 10° N of the Ladakh Intrusives at the time of the collision compared with the same paleoposition of the eurasiatic margin implies a convergence of the order of 1500 km by closure of a marginal basin. An upper bound of the amount of crustal shortening within the Pamir block has been estimated at 200–300 km [18,28], which represents the contribution of folding and thrusting. The very low displacement in latitude of the Tajik basin leads to an estimate of the sum of the effects of underthrusting and lateral extrusion of 700 to 800 km. The total amount of shortening in this western part of the Himalayan collision zone can be estimated to be about 2500 km. The collision of India is now still continuing at a convergence rate of about 2 cm/yr [21,29] and the recent and present day motion of the Tajik basin must be longitudinal to the west.

In the eastern part of the collision zone at the latitude of Lhasa (South Tibet), available data confirm that intracontinental convergence after collision amounts to 2000 km [2,3,30]. This shortening is mainly accounted for by sideway expulsion of blocks [6,7]. South of the Lhasa block, about 500 km of continental crust have disappeared through continental subduction [4]. Consequently, the total amount of shortening of Eurasian margin can also be estimated to be at about 2500 km. The amounts of shortening of the eurasian margin in the western and eastern parts of the collision zone are of comparable amplitude but the different mechanisms of shortening play different roles in the two regions.

It is not possible to end this discussion without pointing out other paleomagnetic results strongly related to this study. Bazhenov and Burtman [9] and Bazhenov et al. [31] have presented results from Upper Eocene to Lower Miocene red beds and grey siltstones sampled in the Tajik basin near Darvaz, in the Peter the First Range and in the Transalai. A total of 157 samples were subjected to thermal and alternating field demagnetization up to 400°C and 150 mT. Hematite has been recognized as the main magnetization carrier and thermal demagnetization increases clustering. Characteristic directions of magnetization have been isolated. Both normal and reversed polarity components have been evidenced with nearly antiparallel directions. A positive fold test has been demonstrated in one locality (Chil'dara) and the magnetizations are considered as primary. These results are summarized in Table 3.

Within the five localities, inclination ranges between 29° and 41° and paleolatitude between 15° and 23° N. This paleolatitude lies far to the south of the paleoposition of the Cretaceous red beds at 36° N. The paleolatitudes are apparently not reconcilable because the positive fold test obtained in the Cretaceous Red Beds determines a firm upper limit of the magnetization at the first Eocene–

Table 3

Results of study of Paleogene rocks [9,31]

Sites	Long.	Lat.	Age	D	I	k	α_{95}	Paleolat.
South Darvaz	37.9	70.1	Late Oligocene–Early Miocene	305	29	11	7	15
Chil'dara	38.8	70.3	Late Eocene–Oligocene–Early Miocene					
Tildara 1				354	30	19	8	16
Tildara 2				349	30	7	15	16
Obikhindou *				325	52			
Khipshun	38.8	70.9	Oligocene	329	40	12	7.5	22.5
Kyzylart	39.4	73.3	Eocene–Oligocene Early Miocene	43	41	11	8.3	23
Khatyn–Kanish	39.4	73.4	Late Eocene–Oligocene	35	37	14	7.6	21.5

* Considered in the mean direction in the original paper [31] and in [10], leading to $D = 346$, $I = 40$, Paleolat. = 22.5, but rejected as abnormal in [9].

Oligocene phase of folding in the Tajik basin. How to remagnetize Cretaceous Red Beds just before the first phase of collision and folding at about 40 Ma at 36° N, and subsequently to magnetize overlaying Tertiary sediments at 23° N.

A similar contradiction has been observed in South Tibet. The lower to middle Cretaceous Takena formation has given a paleolatitude of 20° N [1,2], and 14° N [3], deduced from a prefolding magnetization acquired during the Long Cretaceous Normal Superchron. The paleolatitude of the overlying tertiary Lingzizong formations, sampled in the same area has been found to be 12° N [2,3]. In the same way as we did for the red beds of the Tajik basin, the age of the remagnetization of the Takena formation should probably be revised and fixed at the first phase of the collision in South Tibet. Nevertheless a discrepancy still exists and the paleolatitude of the Lingzizong formations lies south of the underlying Takena. This difference in paleolatitude induces a gap between South Tibet and Eurasia in paleogeographic reconstructions [32].

Low paleolatitudes have been evidenced from Oligocene formations of northeast Japan [33]. The paleomagnetic directions have been compared with the stable Eurasian craton to constrain the models of opening of the Japan Sea. Low paleolatitudes with a discrepancy of 13° indicate a northward shift of Japan of more than 1000 km with respect to Eurasia since the Oligocene, a conclusion not supported by geological evidence.

For the same period, abnormally low paleolatitudes have been discussed by Westphal et al. [23], using paleomagnetic results from Turkey, Iran and Caucasus which indicate positions far to the south of Eurasia while geological evidences demonstrate positions close to Eurasia.

This discepancy leads to the conclusion that the APWP of Eurasia could not be used to describe the motion of all the parts of Eurasia.

For the Tertiary, available data are from formations located in western Europe (Great Britain and Germany). For Cretaceous, paleomagnetic poles have been obtained from formations located at the westernmost and easternmost end of Eurasia. The problem is to know if the Eurasian plate is rigid enough to allow the use of a unique APWP established at both ends of the plate to determine the paleoposition of all the regions situated in between, especially during periods of rapid motion or continental collision.

A discussion of the paleo-stress trajectories over the West European platform since the Eocene lead Le Pichon et al. (34), to conclude that the Oligocene was dominated by an east–west extension pattern related to relative motion between western Europe and central Eurasia. This relative motion was confirmed by the difference between the plate kinematics deduced from Arctic magnetic lineations and those deduced from North Atlantic magnetic lineations. The extensional relative motion of western Europe and central Eurasia of Oligocene was followed by an equivalent compressional relative motion in Early Miocene. The precision of kinematics is too low for these results to be dicussed quantitatively. Particularly, possible differential extensions related to relative rotations are possible and imply large changes in paleolatitude farther east in Central Asia.

Another boundary has been proposed in the Baikal rift zone by Zonenshain and Savostin [35]. The rift is a narrow seismic belt which extends further east along the Stanovoy range and the Okhotsk Sea, and can be considered as a boundary between Eurasian and Amurian lithospheric plates. This result also is a logical consequence of the indentation model.

Other hypotheses have been put forward to explain low paleolatitudes: (1) Abnormal behaviour of the Earth's Magnetic Field has been demonstrated to exist [36]. (2) The effect of compaction which can produce shallowing of inclinations of variable amplitude following the nature of the formation.

But before proof to the contrary, it is normal to test the interpretation given usually for low paleolatitude as position in low latitude and from this point of view, if it can be admitted that differential motion has been experienced by different parts of non rigid Eurasia, one could find there a first step in the explanation of the Oligocene low paleolatitudes.

6. Conclusions

The study of the Lower Cretaceous sediments of the Tajik basin gives a clear characteristic direction of magnetization defined by $D = 350°$, $I = 55.5$, $\alpha_{95} = 4°$, carried by hematite. This magneti-

zation predates the Late Eocene–Early Oligocene phase of folding of the Tajik basin, as indicated by a positive fold test.

A weak reverse component of high blocking temperature, also present in several specimens, is interpreted to be primary and the characteristic direction of magnetization to be a late remagnetization acquired just before the folding of the sediments related to the continental collision.

The comparison of the paleolatitude of Dushanbe, computed from the direction of magnetization and the paleolatitude expected from the APWP of Eurasia, gives an estimate of the post-collision continental shortening north of the Tajik basin, which does not exceed a few hundred kilometers. An anticlockwise rotation of 26° is in good agrement with the indentation model.

A paleogeographic reconstruction at 45 Ma, including the new data on the Tadjik basin, proposes a general array for the Indian and Eurasian margin in the western Himalayas, which is coherent with the scheme established farther east in South Tibet. These reconstruction also indicates the probable existence of a residual ocean floored marginal basin, north of the Ladakh, during that period.

The paleomagnetic data presented by Soviet workers for the overlaying Tertiary sediments of the Tadjik basin contradict our results, and a review of "abnormal" low paleolatitudes found in the Tethyan area indicates that it is not an isolated problem.

This results could possibly be explained by a loss of rigidity of the Eurasian plate during Tertiary and emphasizes the problem of the lack of data for the central part of Eurasia.

Acknowledgements

We thank F. Cornet for his efficient help in the field work and M. Labbeye for his help in laboratory measurements. We had fruitful discussions with J. Besse and with A. Khramov. This is contribution E.N.S. Geology 112.

References

1 M. Westphal, J.P. Pozzi, Y.X. Zhou, L.S. Xing and X.Y. Chen, Paleomagnetic data about southern Tibet, The Cretaceous formations of the Lhasa block, Geophys. J. R. Astron. Soc. 73, 507–521, 1983.

2 J.P. Pozzi, M. Westphal, Y.X. Zhou, L.S. Xing and X.Y. Chen, Position of the Lhasa block south Tibet, during the late Cretaceous, Nature, 297, 5864, 319–321, 1982.

3 J. Achache, V. Courtillot and Y.X. Zhou, Paleogeographic and tectonic evolution of southern Tibet since middle Cretaceous times, new paleomagnetic data and synthesis, J. Geophys. Res. 89, 10,311–10,339, 1984.

4 J. Besse, V. Courtillot, J.P. Pozzi, M. Westphal and Y.X. Zhou, Paleomagnetic estimates of crustal shortening in the Himalayan thrusts and Zang bo suture, Nature, 311, 5987, 621–626, 1984.

5 J.P. Pozzi, M. Westphal, J. Girardeau, J. Besse, Y.X. Zhou, X.Y. Chen and L. Xing, Paleomagnetism of the Xigaze ophiolites and flysch (South Tibet). Latitude and direction of spreading, Earth Planet. Sci. Lett. 70, 383–394, 1984.

6 P. Tapponnier, G. Peltzer, A.Y. Le Dain, R. Armijo and P. Cobbold, Propagating extrusion tectonics in Asia: new insights from simple experiments with plasticine, Geology 10, 611–616, 1982.

7 P. Tapponnier, G. Peltzer and R. Armijo, On the mechanics of the collision between India and Asia, in: Collision Tectonics edited by J.G. Ramsay, M.P. Coward and A. Ries, eds., pp. 115–157, Geol. Soc. London, 1986.

8 V.S. Burtman and G.Z. Gurariy, On the nature of the Pamirs and Tien Shan folded arcs according to paleomagnetic data (in Russian), Geotectonika, 2, 190–195, 1973.

9 M.L. Bazhenov and V.S. Burtman, Formation of the Pamir–Penjab Syntaxis: implications from paleomagnetic investigations of lower Cretaceous and Paleogene Rocks of the Pamirs, in: Contemporary Geoscientific Research in Himalaya, Vol. 1, A.K. Sinha, ed., pp. 71–81, Wadia Inst. Himalayan Geol., Dehra Dune, 1981.

10 C.T. Klootwijk, A summary of paleomagnetic data from extra peninsular Indo-Pakistan and south central Asia: implications for collision tectonics, in: Structural Geology of the Himalaya, P.S. Saklani, ed., pp. 307–360, Today and Tomorrows Printers and Publishers, New Delhi, 1979.

11 J. Besse, Cinématique des plaques et dérive des poles magnétiques. Evolution de la Tethys, collisions continentales et couplage noyau manteau, Thèse, Univ. Paris, 1986.

12 S.A. Zakharov, Problemy geologii Tadjikistana, Akad. Nauk. Tadzh. SSSR, Dushanbe, pp. 33–79, 1964.

13 V.S. Rzhevsky and A.N. Khramov, Palinspastic restorations for the South Tien Shan mountains and Tajik depression based on paleomagnetic data, J. Geodyn. 2, 119–126, 1985.

14 Anonymous, Subdivision of stratified and intrusives rocks of Tadjikistan, Izdat. "Donish", Dushanbe, pp. 1–270, 1976.

15 J. Zlotnicki and F.H. Cornet, A numerical model of earthquake induced piezomagnetic anomalies, J. Geophys. Res. 91, 709–718, 1986.

16 J.L. Kirschvink, The least-squares line and plane analysis of paleomagnetic data, Geophys. J. R. Astron. Soc. 62, 699–718, 1980.

17 M.W. McElhinny, Statistical significance of the fold test in paleomagnetism, Geophys. J. R. Astron. Soc. 8, 338–340, 1964.

18 P. Molnar and P. Tapponnier, Cenozoic tectonics of Asia: effects of a continental collision, Science 189, 419–426, 1975.
19 C.T. Klootwijk, M.L. Sharma, J. Gergan, B. Tirkey, S.K. Shah and V.K. Agarval, The extent of Greater India, II. Paleomagnetic data from the Ladakh Intrusives at Kargil, northwestern Himalya, Earth Planet. Sci. Lett. 44, 47–64, 1979.
20 C.T. Klootwijk and J.W. Pierce, India's and Australia's pole path since late Mesozoic and the India–Asia collision, Nature 282, 605–607, 1979.
21 P. Patriat, J. Segoufin, R. Schlich, J. Goslin, J.M. Auzende, P. Beuzart, J. Bonnin and J.L. Olivet, Les Mouvements relatifs de l'Inde, de l'Afrique et de l'Eurasie, Bull. Soc. Geol. Fr. 7, XXIV, 2, 363–373, 1982.
22 H. Perroud, Relations paleomagnetisme déformations: example de la région de Cabo de Penas (Espagne), C.R. Acad. Sci. Paris Ser. II, 294, 45–48, 1982.
23 M. Westphal, M.L. Bazhenov, J.P. Lauer, D.M. Pechersky and J.C. Sibuet, Paleomagnetic implications and evolutions of the Tethys belt from the Atlantic Ocean to the Pamirs since The Triassic, Tectonophysics 123, 37–82, 1986.
24 E. Irving, Drift of major continental blocks since the Devonian, Nature 270, 304–309, 1977.
25 W. Franck, A. Gansser and V. Trommsdorff, Geological observations in the Ladakh area (Himalaya), a preliminary report. Schweiz. Mineral. Petrogr. Mitt. 57, 89–113, 1977.
26 R.A.K. Tahirkheli, M. Mattauer, F. Proust, P. Tapponnier, The India Asia suture zone in northern Pakistan: synthesis and interpretation of recent data at plate scale, in: Geodynamics of Pakistan, A. Farah and K.A. De Jong, eds., Geol. Surv. Pakistan, Quetta, pp. 125–130, 1979.
27 C.J. Allègre et al., Structure and evolution of the Himalaya Tibet orogenic belt, Nature 307, 17–22, 1984.
28 P. Tapponnier, M. Mattauer, F. Proust and C. Cassaigneau, Mesozoic ophiolites, sutures, and large scale tectonics movements in Afganistan, Earth Planet. Sci. Left. 52, 355–371, 1981.
29 K.N. Kravchenko, Tectonic evolution of the Tien Shan, Pamir and Karakorum, In: Geodynamics of Pakistan, A. Farah and K.A. De Jong, eds., Geol. Surv. Pakistan., Quetta, pp. 25–40, 1979
30 P. Molnar and Wang Ping-Chen, Evidence of large cainozoic crustal shortening of Asia, Nature 273, 218–220, 1978.
31 M.L. Bazhenov, V.S. Burtman and G.Z. Gurariy, Study of the curvature of the Pamirs arc in Paleogene by paleomagnetic method (in Russian), Dokl. Akad. Nauk. SSSR 242(5), 1137–1139, 1978.
32 J. Besse and V. Courtillot, Paleogeographic maps of the continents bordering the Indian ocean since the early Jurassic, J. Geophys. Res. 93, 11,791–11,808, 1988.
33 M.G. Moreau, V. Courtillot and J. Besse, on the possibility of widespread remagnetization of pre-Oligocene rocks from N.E. Japan and the Miocene rotational opening of the Japan Sea, Earth Planet. Sci. Lett. 84, 321–338, 1987.
34 X. Le Pichon, F. Bergerat and M.J. Roulet, Plate kinematics and tectonics leading to the Alpine belt formation; a new analysis, Geol. Soc. Am., Spec. Pap. 218, 111–131, 1988.
35 L.P. Zonenshain and L.A. Savostin, Geodynamics of the Baikal rift zone and plate tectonics of Asia, Tectonophysics 76, 1–45, 1981.
36 R.L. Wilson, Dipole offset. The time average paleomagnetic field over the past 25 My, Geophys. J. R. Astron. Soc. 22, 491–504, 1971.